



Executive Summary: Telco Customer Churn Analysis

The Telco Customer Churn analysis aims to understand the key factors driving customer attrition and identify actionable insights to improve customer retention. This project explores patterns in customer behavior using data visualization, statistical analysis, and predictive modeling techniques.



Overview of the Dataset

- The dataset consists of information on **telecommunication customers**, including demographics, service usage, contract details, and churn status.
- The target variable is **Churn**, indicating whether a customer has left the service (Yes/No).



Churn Rate

- Overall churn rate in the dataset is approximately **26.5%**, meaning **over a quarter** of customers are leaving.
- This highlights a significant business concern and potential revenue impact.



Key Insights & Patterns Identified

1. Demographic Insights

- **Senior Citizens** have a higher churn rate (**~42%**) compared to non-seniors (**~24%**).
- **Single customers** (those without partners or dependents) are more likely to churn.

2. Service Usage Patterns

- Customers with **fiber optic internet** churn at a higher rate (**~41%**) than those with DSL or no internet service.
- Additional services like **online security**, **online backup**, and **tech support** appear to **reduce churn**, indicating bundled service offerings can help retain customers.

3. Contract & Payment Analysis

- **Month-to-month contract customers** churn at **~43%**, compared to **~11%** for **two-year contract holders**.

- Customers using **electronic checks** are more likely to churn (~**33%**) than those using credit cards or bank transfers.
- Prepaid or auto-payment options tend to correlate with **higher retention**.

4. Tenure (Loyalty Indicator)

- Customers with tenure **less than 12 months** show the **highest churn**, especially those in the **first 3 months** of service.
- Retention improves significantly after **24 months**, reinforcing the need for early engagement strategies.



Data Visualizations

The analysis includes a range of informative visualizations:

- **Bar charts** showing churn by internet service type, payment method, contract type, and customer tenure.
- **Pie charts** representing the distribution of churn across various demographics.
- **Histograms and box plots** to explore numerical features like monthly charges and tenure.
- **Correlation heatmaps** and feature importance from machine learning models to identify key churn predictors.



Predictive Modeling

- A classification model (likely Logistic Regression, Random Forest, or XGBoost) was used to predict churn.
- **Model Accuracy:** ~80%
- **Important Features:** Contract type, tenure, monthly charges, and internet service type.



Business Recommendations

1. **Promote longer-term contracts** with discounts or loyalty benefits.
2. **Targeted engagement** for new users during their first 3-6 months.
3. Offer **value-added services** (like tech support) as part of premium packages.
4. Provide **payment incentives** for switching from electronic checks to auto-pay methods.
5. Launch a **senior-friendly support initiative** to cater to older customers with higher churn risk.

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